

PHYSICS 100 · FINAL PROJECT · STANFORD STUDENT OBSERVATORY

V-band Time-Series Photometry of AM CVn

Independent recovery of the superhump and its first harmonic

Sagittarius A*s

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INTRODUCTION

What is an AM CVn star?



Hydrogen-deficient interacting binaries

White dwarf accretes helium-rich matter from hydrogen-poor donor through an accretion disk.



Orbital periods of only ~5 - 65 minutes

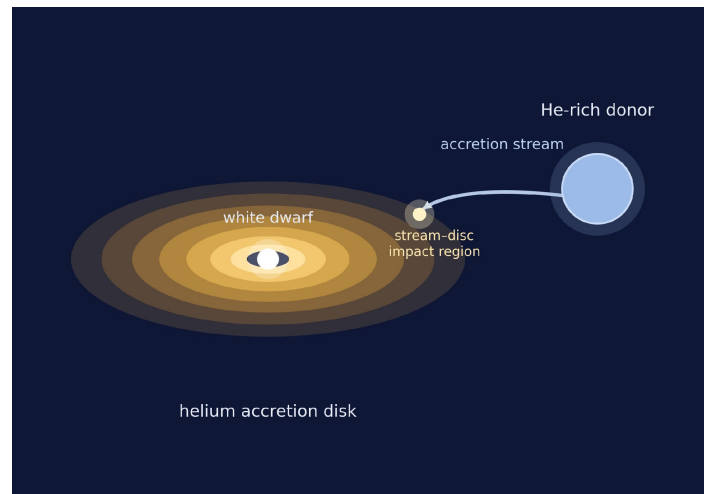
Compact binaries with thermally stable disks at short periods.



Probes of binary evolution and gravity

Angular-momentum loss is driven by gravitational radiation.

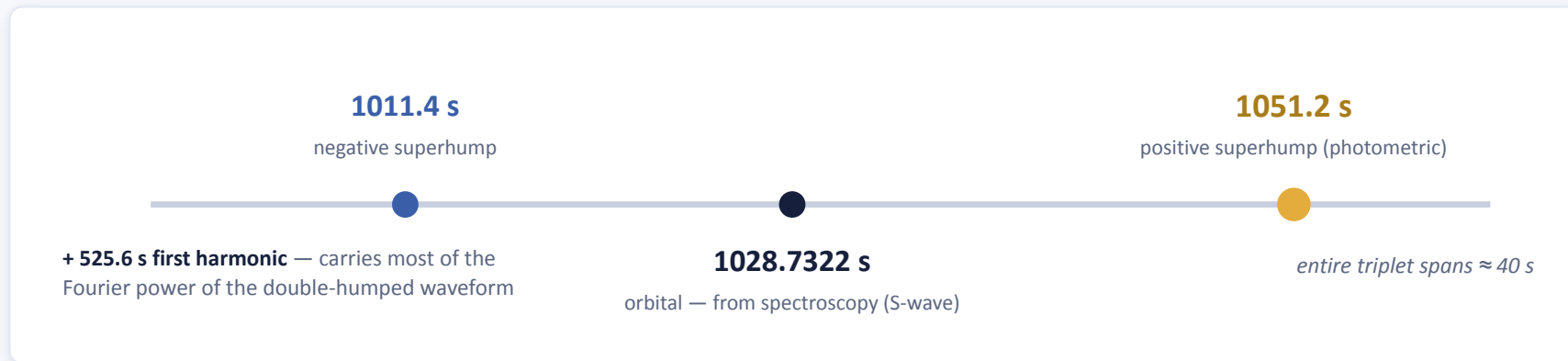
AM CVn is the class prototype: blue, hydrogen-deficient spectrum and optical variability on minute timescales.



Mass flows from the donor through a stream onto a helium disk around the white dwarf.

INTRODUCTION

Three closely spaced clocks



GOAL 1

Recover the known signal

Detect the positive superhump and its first harmonic from a single 111.12-minute night.

GOAL 2

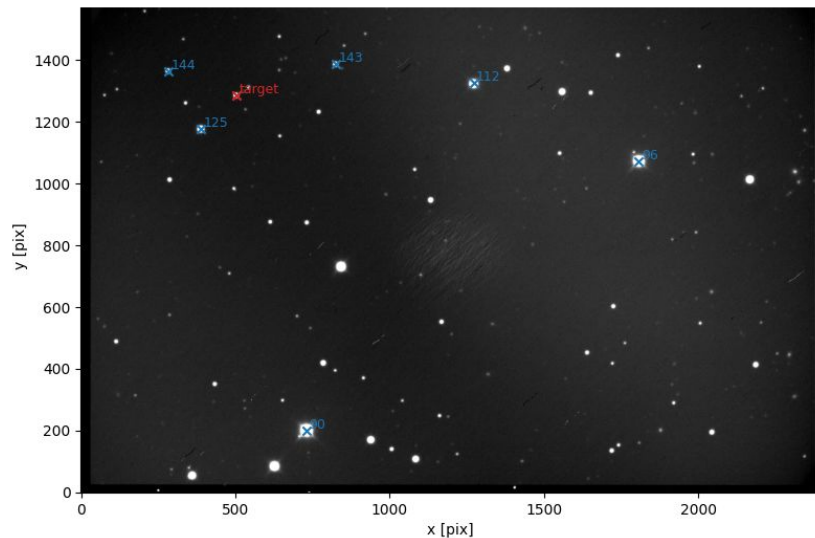
Tighten the periods?

Test whether a free, independent fit can improve the published period constraints.

independent = period fit freely from our data · *conditional* = a literature identification is assumed

OBSERVATIONS

Observing on the Stanford 0.7 m



Stacked V-band field ($N = 138$).

Target in red; six AAVSO comparison stars (sequence X42550AJH, $m < 15$) in blue.

TELESCOPE

0.7 m · Stanford Student Observatory

NIGHT

20 May 2026 · 22:20–00:30 PDT

SETUP

V band · 4×4 binning · 27' × 18' FOV

FRAMES

140 → 138 kept · 35 / 40 / 45 s exposures

last 2 dropped: severe artifacts + low counts

CONDITIONS

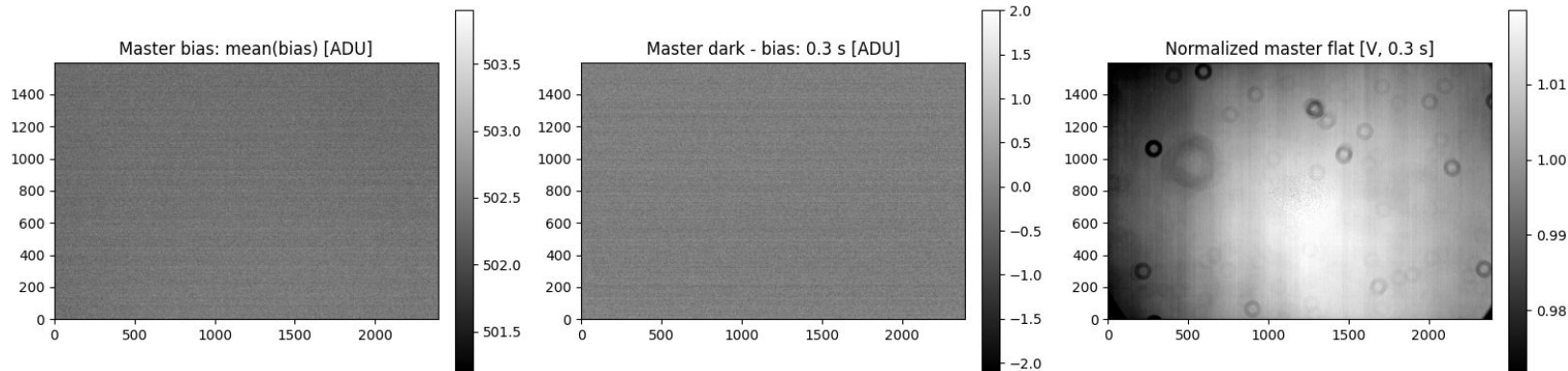
Clear · seeing 5"–8" · altitude 85° → 60°

BASELINE

111.12 min = 6.34 superhump cycles

Raw frames → light curve

$$I_{\text{cal}} = \frac{I_{\text{raw}} - B - D(t_{\text{exp}})}{F}$$



Master bias (10 frames) · master darks (29 frames, matched to 0.3 / 35 / 40 / 45 s) · normalized 0.3 s V master flat (10 frames borrowed from 16 April)

1

Calibrate

bias, matched darks, flats

2

Plate-solve

ASTAP WCS on 138 frames

3

Locate stars

AAVSO coords + centroid
refit (8 px)

4

Photometry

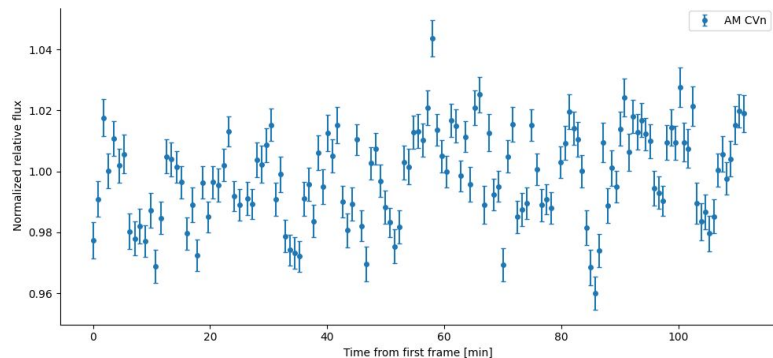
6 px aperture · 12–20 px
sky annuli

5

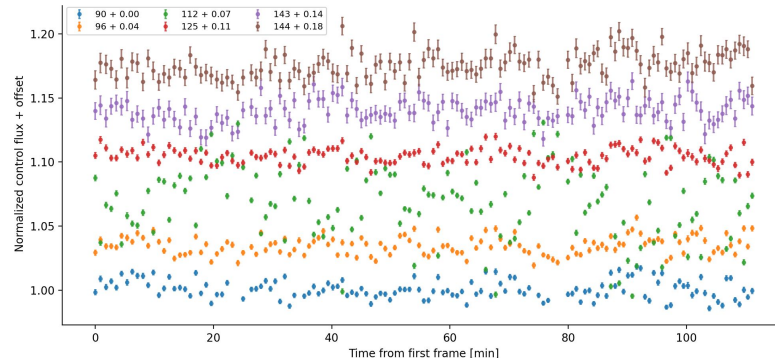
Ensemble

count rates, normalized ÷
6-comp mean

Is the variability from AM CVn?



AM CVn ensemble differential light curve — 138 points over 111.12 min.



Leave-one-out controls: each comparison treated as a pseudo-target. Five sit at ~0.6–1.1% scatter; star 112 at 2.9%. None shows AM CVn-like variability.

Leave-one-out light curves

Standard practice to validate comparison stars

$$\Delta P = 0.40 \text{ s}$$

if exclude noisiest comp (112) — below the 6.7 s period uncertainty → retain full ensemble

Controls are flat

differential photometry does not introduce the signal → variability from AM CVn

A ladder of models

MODEL LADDER

- | | | |
|---|---|-------------|
| 0 | Constant flux
no coherent variability | BASELINE |
| 1 | Free single sine
one sinusoid, period fit freely | INDEPENDENT |
| 2 | 1051.2 s + harmonic
fixed literature superhump model | CONDITIONAL |
| 3 | Beat-constrained proxy
superhump frequency = orbital – precession (≈ 13.4 h; cover 0.14 cycle) | CONDITIONAL |
| 4 | Free double wave ★
fundamental + first harmonic, period free | INDEPENDENT |

FITTING & COMPARISON

$$m(t_i) = c + \sum_{j=1}^M [a_j \sin(2\pi h_j t_i / P_j) + b_j \cos(2\pi h_j t_i / P_j)]$$

$$\text{BIC}_{\text{eff}} = \chi^2 + k_{\text{eff}} \ln N$$

Weighted χ^2 fits; Lomb–Scargle as search diagnostic.
 k_{eff} counts fitted period; BIC as heuristic.

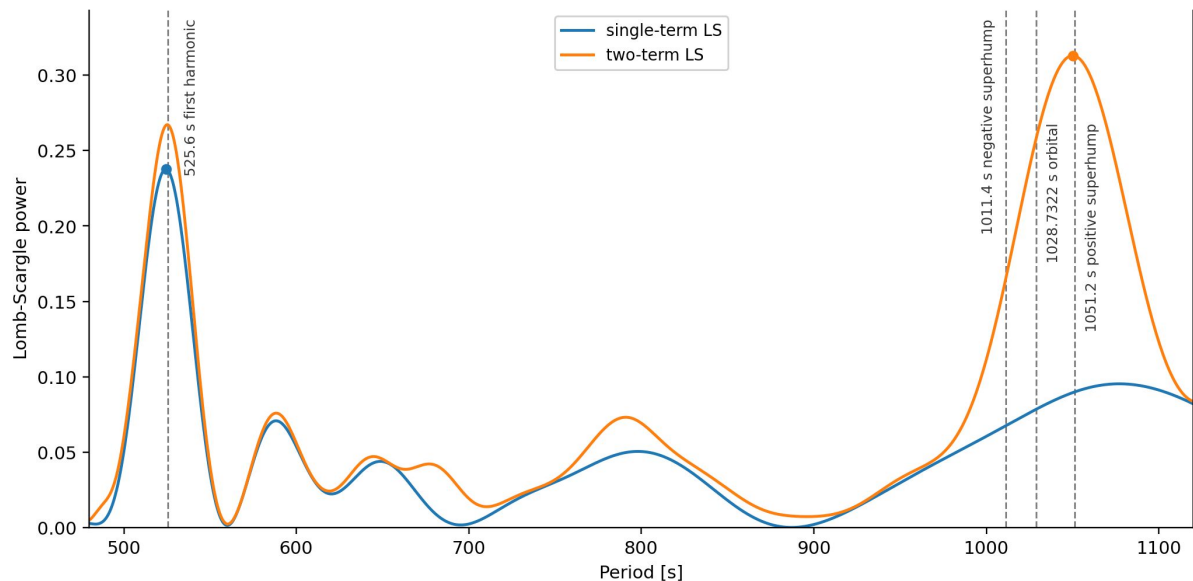
UNCERTAINTIES & STRESS TESTS

Parametric Monte Carlo — synthetic light curves on real time stamps, errors scaled $\times 2.27$ ($\chi^2_{\nu} \approx 5.2$), full search re-run each trial.

Stress tests — half-night splits, $|z| > 4$ clipping, comp-star drops, robust losses, nuisance regressors, time-blocked cross-validation.

RESULTS

Detection — through the first harmonic



Single-term (blue) and two-term (orange) Lomb–Scargle periodograms. Dashed lines: published clocks. The triplet near 1050 s blends into a single peak.

524.97 s

single-term LS peak — false-alarm probability 1.8×10^{-5}

524.35 ± 3.3 s

best free single-sine period (Monte Carlo error)

≈ 1053 s

two-term LS recovers the double-wave fundamental

One sinusoid is enough to detect AM CVn — but recovering the 17.5-min double wave requires a two-term or physically motivated model.

RESULTS

An independent (not tighter...) period

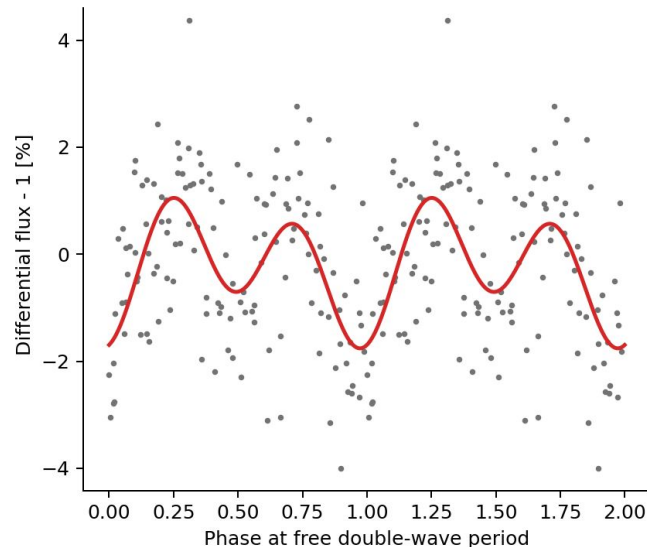
FREE DOUBLE-WAVE PERIOD

1050.05 $\pm$ 6.7 s

within 1σ of the published 1051.2 s positive superhump — not precise enough to improve the literature.

Model	Period (s)	RMS (%)	$\Delta\text{BIC}_{\text{eff}}$
Constant flux	—	1.50	+288.75
Free single sine	524.35	1.31	+65.47
1051.2 s + harmonic (cond.)	1051.20	1.25	-4.74
Beat-constrained proxy (cond.)	1050.03	1.25	0.00
Free double wave ★	1050.05	1.25	0.00

$\Delta\text{BIC}_{\text{eff}}$ is relative to the free double wave. The fixed template is strongest conditionally on the literature period; the free fit is the independent measurement.



Folded at the free 1050.05 s period: double-humped — first-harmonic amplitude 1.00% exceeds the fundamental's 0.58%, matching published waveform.

Stable to outlier clipping (Δ 0.08 s) and comparison choice (Δ 0.40 s) — but half-night fits differ by 16.4 s ($\approx 2.45\sigma$): intrinsic waveform variability.

What one short baseline can & can't do



Superhump recovered independently

Harmonic at 524.35 ± 3.3 s (FAP 1.8×10^{-5}); free double-wave period 1050.05 ± 6.7 s — within 1σ of the published 1051.2 s.



Waveform matches literature

Double-humped, with the first-harmonic amplitude (1.00%) exceeding the fundamental (0.58%)



No tighter constraints

$\delta P \approx P^2/T \approx 165$ s cannot split the 17–40 s triplet; we cover 0.14 precession cycles; half-night drift of 16.4 s $\Rightarrow$ baseline + intrinsic waveform variability limits



NEXT STEPS

≥ 13.4 h baseline

span one full disc-precession cycle to resolve the superhump–orbital triplet photometrically.

Time-resolved spectroscopy

catch the S-wave from the stream–disc impact directly.

Thank you — questions?